

FRANCISCO BETANCOURT

Senior Embedded & Systems Software Engineer

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PROFESSIONAL SUMMARY

Senior Embedded Software Engineer with 12+ years of experience architecting production-grade firmware, safety-critical automotive ECUs, autonomous edge systems, and connected IoT platforms. Proven track record in C/C++ bare-metal and RTOS (FreeRTOS/Zephyr) environments, low-level MCU drivers (NXP S32K, STM32, Nordic nRF52, ESP32), automotive diagnostics (UDS/CAN), power management/BMS optimization, and Embedded Linux HIL test automation. Strong background in control systems (PID, IMU/GNSS sensor integration) and autonomous robotics concepts (ROS 2, LiDAR fundamentals).

- **Domain Specializations:** High-reliability roles across **Autonomous Vehicles & Mobility, Robotics & Autonomous Systems, Automotive Power & Battery Management Systems (BMS)**, and **Connected Industrial IoT Edge**.

CORE TECHNICAL COMPETENCIES

- **Languages & Core Firmware:** Embedded C, C++11/17, Assembly, Python, Bash, MISRA-C Compliance, Static Analysis (Helix QAC).
- **Automotive ECUs & Power/BMS:** NXP S32K118, UDS Diagnostics (ISO 14229), CAN, LIN, Functional Safety (SCST / ISO 26262 concepts), Battery Power Optimization, Duty Cycling & Low-Power Sleep Modes.
- **Real-Time Operating Systems & Architecture:** FreeRTOS, Zephyr RTOS, Startup Code, MPU Configuration, Memory Footprint Optimization, Multi-Threading, Inter-Process Communication (Queues/Semaphores).
- **Robotics, Control & Autonomous Edge:** PID Closed-Loop Control, IMU Motion Processing (ICM-20607 / MPU6050), GNSS/GPS NMEA Parsing, ROS 2 (rclpy/rclcpp fundamentals), LiDAR Concepts, Sensor Fusion.
- **Wireless, Telematics & IoT Protocols:** Bluetooth Low Energy (BLE GAP/GATT), Wi-Fi (ESP-IDF / TCP/IP), Cellular 3G/4G, MQTT, WebSockets, I2C, SPI, UART, I2S, USB, MDB/DEX.
- **Embedded Linux & HIL Automation:** Embedded Linux BSP/Kernel debugging, Python/Qt Automated HIL Test Benches, gdb, TRACE32/Lauterbach, Segger J-Link, ST-Link, Logic Analyzers (PulseView/sigrok).

PROFESSIONAL EXPERIENCE

Senior Embedded Software Engineer

Joyson Safety Systems | San Nicolás de los Garza, Mexico | 01/2023 – 01/2026 * Developed safety-critical embedded firmware in **C** for automotive ECUs based on **NXP S32K118** microcontrollers across integration, testing, and production validation. * Designed low-level platform architecture including interrupt vector handling, Memory Protection Unit (MPU) configuration, and hardware startup code shared across ECU application baselines. * Implemented **SCST (Self-Test)** mechanisms to fulfill functional safety requirements during system boot and runtime execution in automotive safety systems. * Designed and maintained **UDS Diagnostic Services (ISO 14229)**, including DID management, Security Access algorithms, and fault memory handling for development and production diagnostics. * Utilized vector toolchains (CANoe, Vehicle Spy, TRACE32) for bus monitoring, automated testing, and low-level hardware-in-the-loop debugging.

Senior Embedded & IoT Firmware Engineer

Vend-In / Telematics Solutions | Monterrey, Mexico | 01/2013 – 01/2023 * Architected multi-radio connected telematics devices combining **Nordic nRF52840 (BLE)**, **PIC32MZ**, and **Cellular/Wi-Fi (ESP32)** for large-scale industrial IoT deployments. * Implemented low-power state machines and BLE GATT services (nRF5 SDK / FreeRTOS), reducing standby current consumption and optimizing battery life for field-deployed devices. * Developed network communication pipelines over Wi-Fi and Cellular using **TCP/IP, MQTT, and WebSockets**, enabling remote telemetry, over-the-air (OTA) firmware updates, and cloud synchronization. * Integrated peripheral interfaces (**I2C, SPI, UART, I2S**) for IMU motion sensing, DEX telemetry data extraction, and dynamic display controllers. * Built custom **Python/Qt HIL automated test benches** on Raspberry Pi Linux, automating firmware validation, hardware signal injection, and regression testing prior to production release.

Embedded Firmware & Systems Engineer

Discapel | San Nicolás de los Garza, Mexico | 11/2011 – 12/2012 * Developed embedded firmware and Linux-based control software for medical, commercial audio, and security devices. * Implemented real-time event-driven audio pipelines using **FreeRTOS on ESP32-A1S** over **I2S**, deploying solutions across commercial retail networks (7-Eleven). * Integrated **NFC/RFID (PN532 / MIFARE)** and **GNSS/GPS (ATGM336H)** modules over UART/I2C, implementing real-time NMEA sentence parsing and location tracking.

Control & Automation Research Assistant

Facultad de Ciencias Físico-Matemáticas (FCFM - UANL) | 10/2008 – 09/2010 * Designed and validated **Simulink/MATLAB-based real-time PID controllers** for dynamic propeller systems and optical measurement hardware. * Co-authored research patents and automated measurement software integrating high-precision motion control and sensor data acquisition.

EDUCATION & CONTINUOUS LEARNING

- **B.S. in Computer Science** — *Universidad Autónoma de Nuevo León (UANL)* (2005 – 2010)
- **Ongoing Specializations:** Advanced ROS 2 Robotics Architecture, Autonomous Vehicle Sensor Fusion, Zephyr RTOS Industrial Patterns.